

Easy (10 questions)

1.

What does subnetting do?

- A) Combines multiple networks into one
- B) Divides a network into smaller networks
- C) Encrypts IP packets
- D) Translates IP to MAC addresses

Answer: B

Explanation: Subnetting divides a large network into smaller subnetworks.

2.

What is the default subnet mask for Class C?

- A) 255.0.0.0
- B) 255.255.0.0
- C) 255.255.255.0
- D) 255.255.255.255

Answer: C

Explanation: Class C default subnet mask is 255.255.255.0.

3.

How many hosts are available in a subnet with a /30 mask?

- A) 2
- B) 4
- C) 6
- D) 8

Answer: A

Explanation: /30 has $2^{(32-30)} - 2 = 2$ hosts.

4.

What is the purpose of a subnet mask?

- A) Encrypt IP addresses
- B) Identify network and host portions of an IP address
- C) Route packets between networks
- D) Translate domain names

Answer: B

Explanation: Subnet mask separates the network and host parts.

5.

Which part of the IP address is used to identify a subnet?

- A) Network bits only
- B) Host bits only
- C) Borrowed bits from host portion
- D) Broadcast bits

Answer: C

Explanation: Subnetting borrows bits from the host portion for subnetting.

6.

What is VLSM?

- A) Using the same subnet mask for all subnets
- B) Using different subnet masks in the same network
- C) Assigning static IPs
- D) Translating IP to MAC addresses

Answer: B

Explanation: VLSM allows different subnet masks within the same network.

7.

In subnetting, why subtract 2 from the number of host bits?

- A) For routing and DNS

- B) For network and broadcast addresses
- C) For encryption keys
- D) For packet checksums

Answer: B

Explanation: Network and broadcast addresses cannot be assigned to hosts.

8.

What is the subnet mask for /27?

- A) 255.255.255.128
- B) 255.255.255.224
- C) 255.255.255.240
- D) 255.255.255.248

Answer: B

Explanation: /27 corresponds to 255.255.255.224.

9.

How many subnets can you create by borrowing 3 bits?

- A) 3
- B) 6
- C) 8
- D) 16

Answer: C

Explanation: $2^3 = 8$ subnets.

10.

Which notation represents an IP address with subnet mask?

- A) Dotted decimal
- B) CIDR
- C) Hexadecimal
- D) Binary

Answer: B

Explanation: CIDR notation uses slash prefix (e.g., /24).

Medium (30 questions)

11.

How many hosts are possible on a subnet with mask 255.255.255.192?

- A) 62
- B) 64
- C) 126
- D) 128

Answer: A

Explanation: 255.255.255.192 = /26, so hosts = $2^{(32-26)} - 2 = 62$.

12.

What is the wildcard mask for subnet mask 255.255.255.248?

- A) 0.0.0.7
- B) 0.0.0.8
- C) 0.0.0.15
- D) 0.0.0.31

Answer: A

Explanation: Wildcard mask = 255 - subnet mask = 0.0.0.7.

13.

Which of these is a valid subnet for 192.168.1.0/27?

- A) 192.168.1.32
- B) 192.168.1.40
- C) 192.168.1.50
- D) 192.168.1.200

Answer: A

Explanation: /27 subnets increase by 32, so 192.168.1.32 is valid subnet start.

14.

How many bits are left for hosts in a /28 subnet?

- A) 4
- B) 8
- C) 16
- D) 32

Answer: A

Explanation: /28 means 28 network bits, so 4 host bits ($32-28=4$).

15.

If a subnet mask is 255.255.255.0, what is the prefix length?

- A) /8
- B) /16
- C) /24
- D) /32

Answer: C

Explanation: 255.255.255.0 corresponds to /24.

16.

In VLSM, what is the strategy for subnet allocation?

- A) Allocate smallest subnet first
- B) Allocate largest subnet first
- C) Allocate random subnet

D) Allocate based on MAC address

Answer: B

Explanation: Allocate largest subnets first to avoid fragmentation.

17.

Which subnet mask provides 14 usable hosts?

A) /27

B) /28

C) /29

D) /30

Answer: B

Explanation: /28 = 16 addresses - 2 = 14 hosts.

18.

What is the broadcast address for 192.168.10.64/26?

A) 192.168.10.65

B) 192.168.10.127

C) 192.168.10.63

D) 192.168.10.128

Answer: B

Explanation: /26 subnets have a block size of 64, broadcast is last address (64+63=127).

19.

How many hosts can be supported on a subnet with /16 mask?

A) 254

B) 256

C) 65534

D) 65536

Answer: C

Explanation: /16 means 16 host bits, $2^{16}-2=65534$ hosts.

20.

What does the “host portion” of an IP address specify?

- A) Network address
- B) Device within subnet
- C) Gateway address
- D) Broadcast domain

Answer: B

Explanation: Host bits identify devices on the subnet.

21.

What is the first valid host address in the 192.168.5.0/29 subnet?

- A) 192.168.5.0
- B) 192.168.5.1
- C) 192.168.5.7
- D) 192.168.5.8

Answer: B

Explanation: Network address is 192.168.5.0, first host is .1.

22.

What is the subnet mask in binary for /30?

- A) 11111111.11111111.11111111.11111100
- B) 11111111.11111111.11111111.11110000
- C) 11111111.11111111.11111111.11111111
- D) 11111111.11111111.11111111.11111000

Answer: A

23.

Which protocol supports VLSM?

- A) RIP version 1

- B) RIP version 2
- C) ARP
- D) ICMP

Answer: B

Explanation: RIP v2 supports CIDR and VLSM, RIP v1 does not.

24.

What is the maximum number of subnets with a /24 network and /28 subnet mask?

- A) 4
- B) 8
- C) 16
- D) 32

Answer: C

Explanation: /24 to /28 means 4 bits borrowed, so $2^4=16$ subnets.

25.

Why is VLSM better than fixed-length subnetting?

- A) It simplifies routing
- B) It allows different sized subnets
- C) It encrypts traffic
- D) It assigns IP addresses randomly

Answer: B

26.

What is the subnet address for IP 192.168.1.130 with mask 255.255.255.192?

- A) 192.168.1.64
- B) 192.168.1.128
- C) 192.168.1.192
- D) 192.168.1.255

Answer: B

Explanation: Block size is 64, 128–191 covers 130.

27.

In binary, what does the subnet mask 255.255.255.0 look like?

- A) 11111111.11111111.11111111.00000000
- B) 11111111.11111111.00000000.00000000
- C) 11111111.00000000.00000000.00000000
- D) 00000000.00000000.00000000.00000000

Answer: A

28.

What is the maximum hosts in a /29 subnet?

- A) 4
- B) 6
- C) 8
- D) 14

Answer: B

Explanation: /29 = 3 host bits = $2^3 - 2 = 6$ hosts.

29.

Which subnet mask corresponds to 255.255.255.252?

- A) /29
- B) /30
- C) /31
- D) /32

Answer: B

Explanation: 255.255.255.252 = /30

30.

The process of dividing a subnet into smaller subnets is called:

- A) Supernetting
- B) VLSM
- C) Aggregation
- D) DHCP

Answer: B

Hard (10 questions)

31.

Given 172.16.0.0/16, you need 6 subnets each with at least 200 hosts. What is the subnet mask?

- A) /24
- B) /25
- C) /26
- D) /27

Answer: A

Explanation: For 200 hosts, need 8 host bits ($256 \text{ addresses} - 2 = 254 \text{ hosts}$), so /24 mask.

32.

What is the broadcast address of 10.10.10.128/26?

- A) 10.10.10.191
- B) 10.10.10.127
- C) 10.10.10.129
- D) 10.10.10.255

Answer: A

33.

Which subnet mask would allow 1000 hosts per subnet?

- A) /22
- B) /23
- C) /24
- D) /25

Answer: A

Explanation: /22 = 10 bits host $\Rightarrow 2^{10} - 2 = 1022$ hosts.

34.

If you have network 192.168.0.0/23, how many subnets with /25 can you create?

- A) 1
- B) 2
- C) 4
- D) 8

Answer: C

Explanation: /23 to /25 means 2 bits borrowed $\Rightarrow 2^2=4$ subnets.

35.

For IP 192.168.15.130 with subnet mask 255.255.255.224, what is the subnet ID?

- A) 192.168.15.128
- B) 192.168.15.96
- C) 192.168.15.160
- D) 192.168.15.224

Answer: A

36.

What is the first valid host of 172.16.100.64/26?

- A) 172.16.100.64
- B) 172.16.100.65

C) 172.16.100.126

D) 172.16.100.127

Answer: B

37.

In VLSM, what is the next subnet if you allocate 192.168.10.0/26 first?

A) 192.168.10.64/26

B) 192.168.10.128/26

C) 192.168.10.192/26

D) 192.168.10.0/27

Answer: A

38.

Which subnet mask allows for exactly 254 hosts?

A) /23

B) /24

C) /25

D) /26

Answer: B

Explanation: /24 $\Rightarrow$ $256 - 2 = 254$ hosts.

39.

What is the decimal block size for subnet mask 255.255.255.240?

A) 8

B) 16

C) 32

D) 64

Answer: B

Explanation: Block size = $256 - 240 = 16$.

40.

Which of the following IPs belongs to the subnet 10.0.0.0/22?

A) 10.0.0.255

B) 10.0.3.1

C) 10.0.4.1

D) 10.0.7.255

Answer: B

Explanation: /22 block covers 10.0.0.0 - 10.0.3.255